PROJECT PROPOSAL

Loan Approval Prediction

A machine learning model that predicts whether a loan application will be approved.

Dataset: Kaggle Loan Prediction **Prepared by:** TEAM OHS – HRMMS [Ballego, Buenafe,Diesto,Gloria]
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614	11	68.7%	7
loan applications	input features	approved (31.3% rejected)	columns with missing values

01 Dataset description

614 past loan applications from Kaggle. Each row is one applicant, described by 11 details (features), plus the outcome we want to predict, **Loan_Status** (Y = approved, N = rejected). Loan_ID is only an ID, so it is not used.

Applicant: Gender, Married, Dependents, Education, Self_Employed
Financial: ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term (months), Credit_History (1 = good, 0 = not)
Property: Property_Area

Credit history matters: among applicants with a recorded history, 79.6% of those with a good one were approved vs 7.9% of those without.
Missing data: 7 columns have gaps, most in Credit_History (50 rows), so they must be filled in before modelling.
Uneven incomes: applicant income runs from 150 to 81,000 (median about 3,800), so a log transform will be tested to tame very large values.

02 Problem statement

Goal: given a loan applicant's details (family and job situation, income, loan amount and term, credit history, property area), predict whether the application is **approved (Y) or rejected (N)**. This is a binary classification problem.

Use: a decision-support tool that helps a lender screen applications faster and flag unclear cases for a person to review. It learns from past decisions and does not replace the loan officer.

Fairness check: Gender and Married are sensitive details. We will compare results across these groups and remove the features if they add little predictive value. With only 614 rows, results may vary between data splits.

03 Chosen metric

Macro F1 with stratified 5-fold cross-validation

Why: accuracy alone misleads, since always guessing "approved" is right about 69% of the time. F1 balances precision (how many predicted approvals are correct) and recall (how many real approvals are found). Macro F1 averages the F1 of the approved and rejected classes, so both counts equally.

Also reported: precision and recall per class, ROC-AUC and a confusion matrix. We assume approving a loan that should be rejected costs more than the reverse, so we will check recall on the rejected class (how many real rejections the model catches).

04 Baseline idea

1. **Majority class:** always predict "approved". Accuracy is 68.7%, but macro F1 is only about 0.41, the score to beat.
2. **Simple rule:** approve only if Credit_History = 1, to see how far one feature alone gets us.
3. **Logistic Regression** (main baseline): fill gaps (median for numbers, most common value for categories), turn categories into numbers (one-hot), scale values, log-transform income and loan amount, and balance the classes. All steps are fitted inside each cross-validation fold to avoid data leakage.